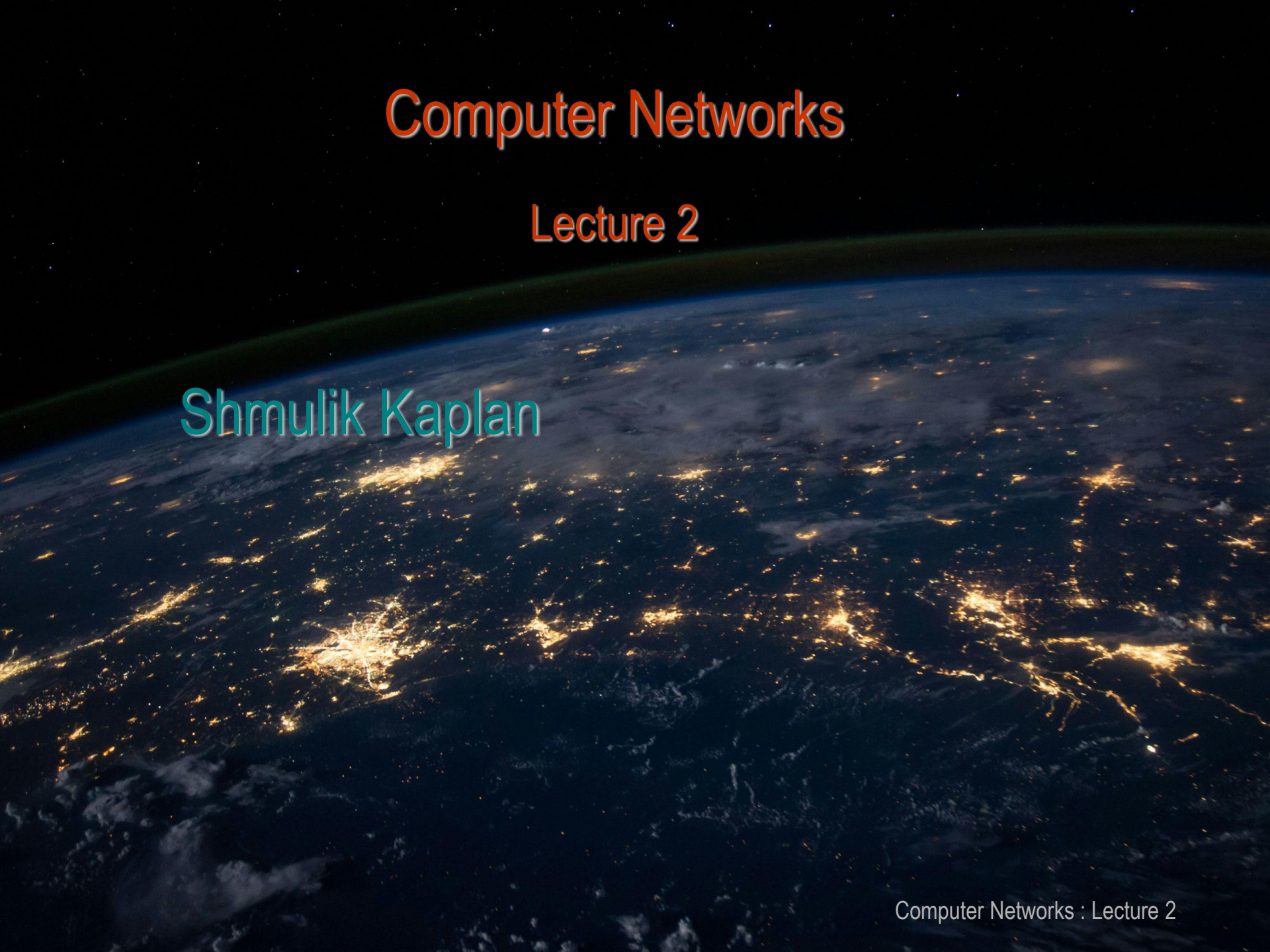


Computer Networks

Lecture 2

Shmulik Kaplan

Today

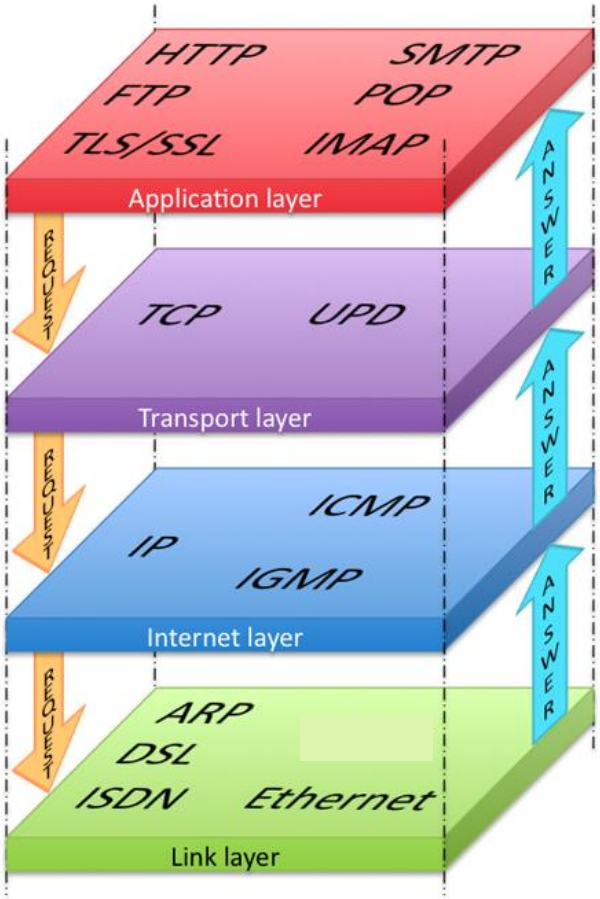
- Last week Review
- Introducing the 5 Layers of the internet protocol – cont.
  - Application Layer
  - Transport Layer
  - Network Layer
  - Link Layer
  - Physical Layer
- Delays in packet switching
- Packet loss
- Virtual Circuit
- Creating a network-application



Computer Networks

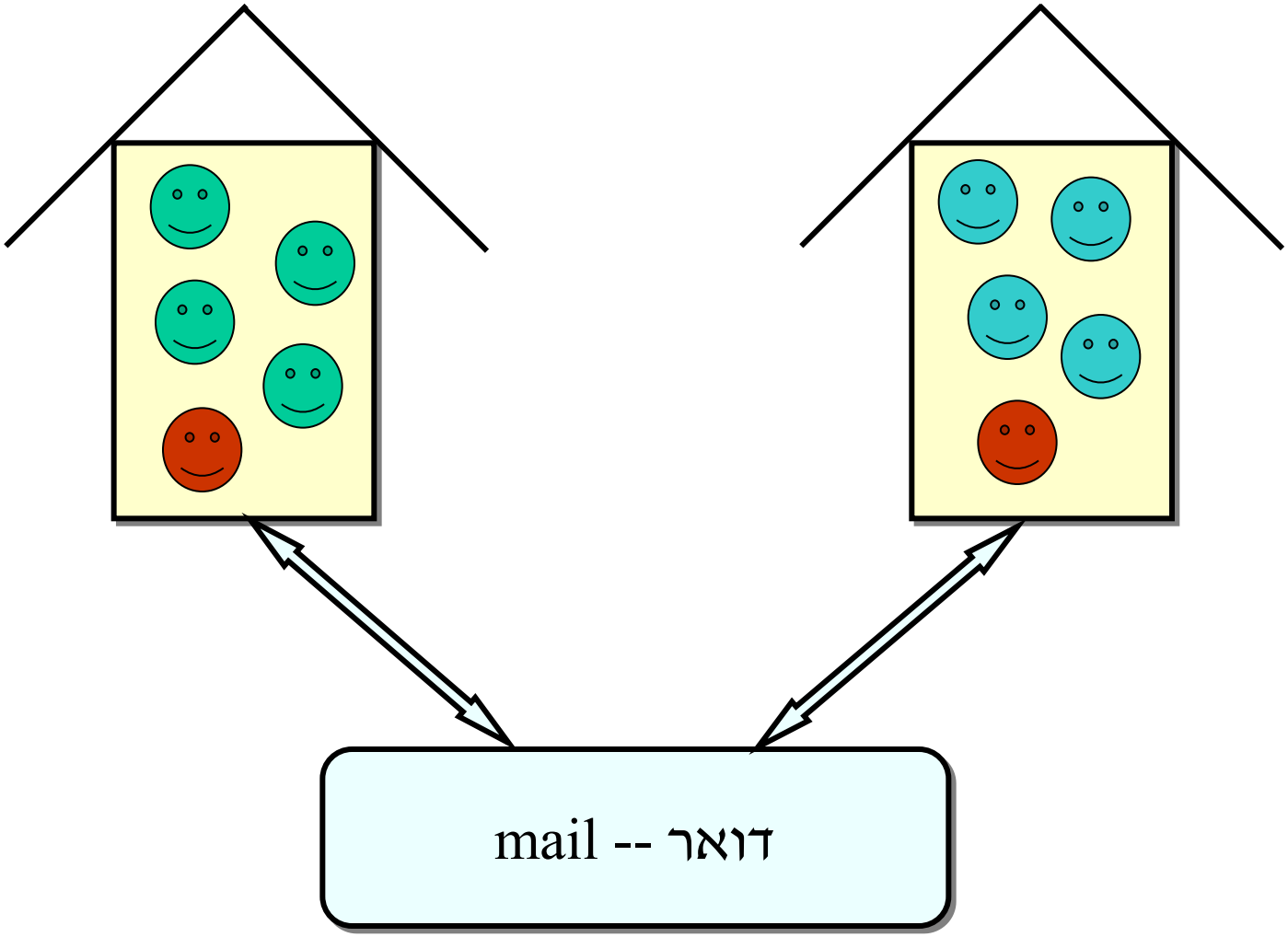
Shmulik Kaplan

- Internet protocol stack
- Packet switching
- Packet size
- Delay in packet-switched networks
- Virtual circuit networks
- Warriors of the Net: The movie



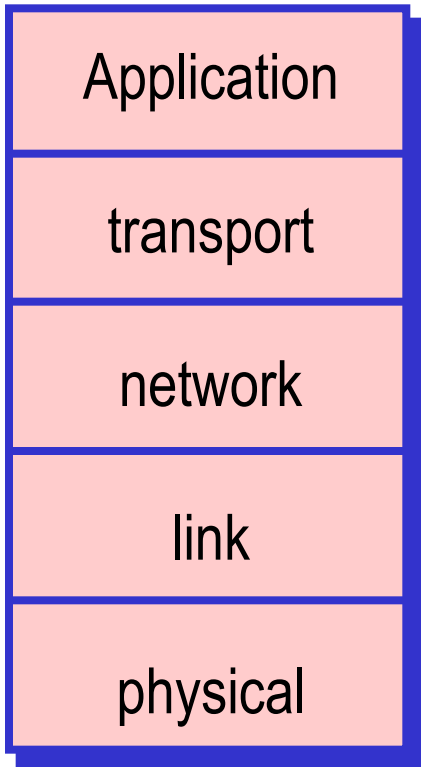
Layering

Example from Lecture 1



Internet protocol stack

- Application: supporting network applications
  - FTP, SMTP, HTTP
- Transport: application to app. data transfer
  - TCP, UDP
- Network: routing of datagrams from source to destination (host to host)
  - IP, routing protocols
- Link: data transfer between neighboring network elements
  - PPP, Ethernet
- Physical: bits “on the wire”

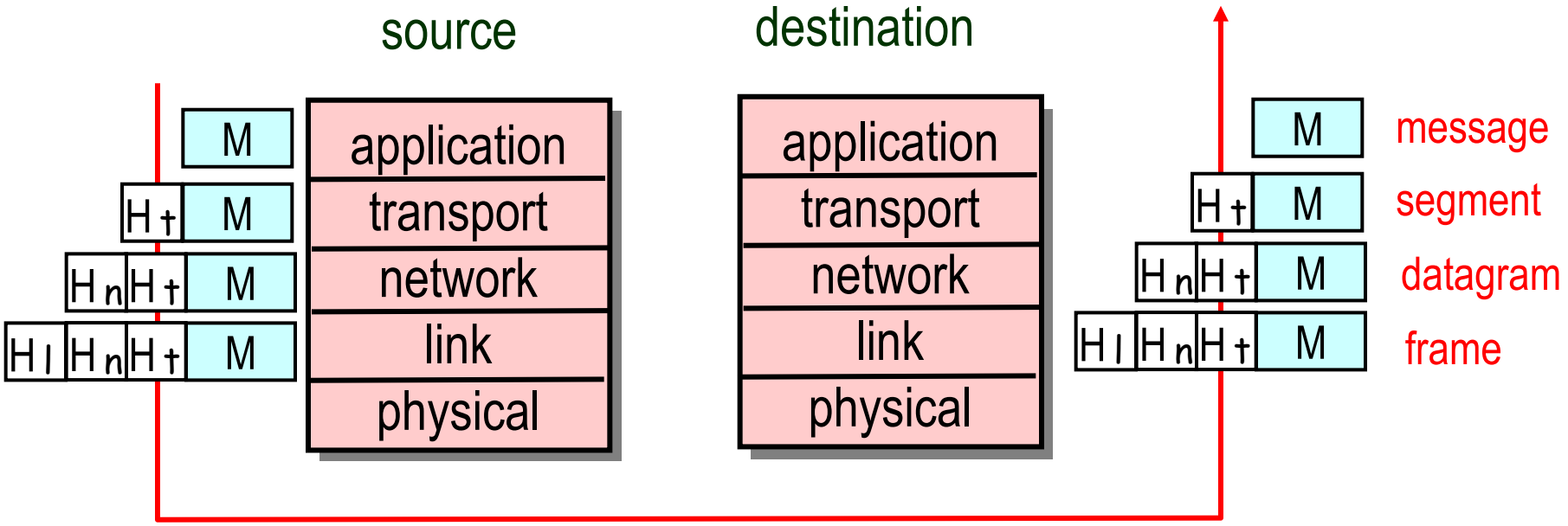


Today

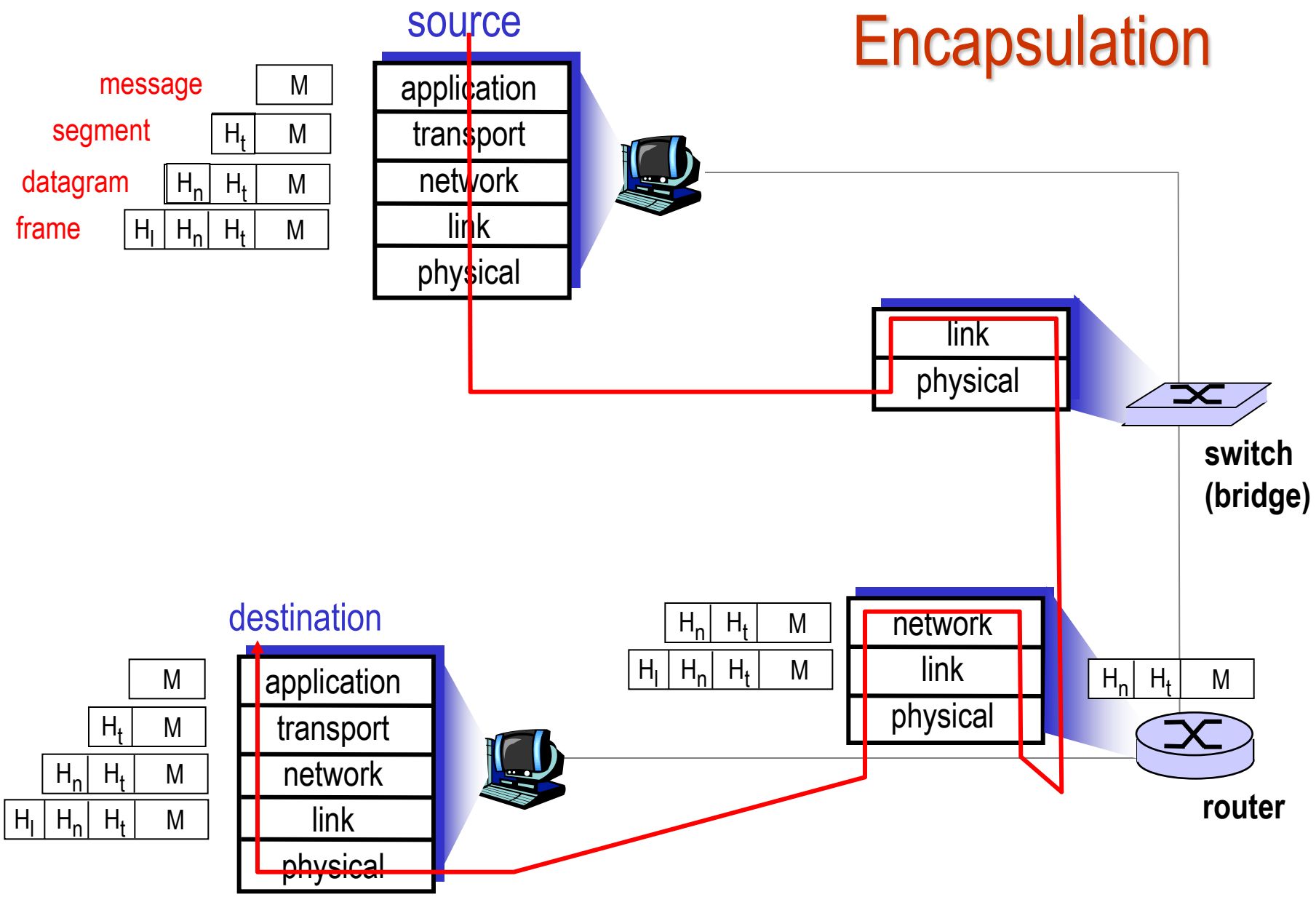


- Last week Review
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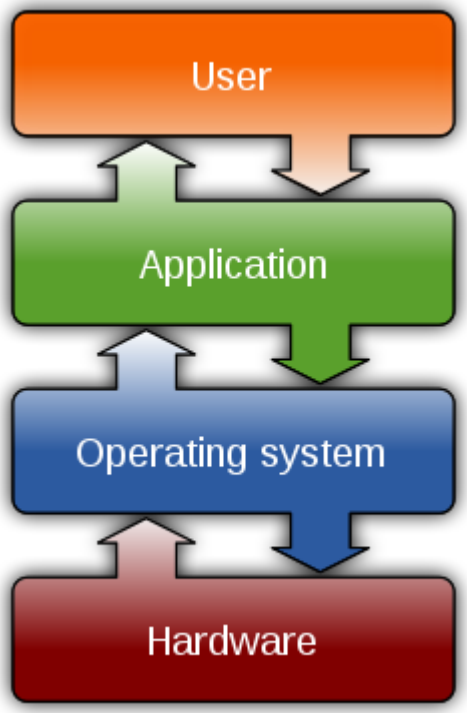
Protocol layering and data



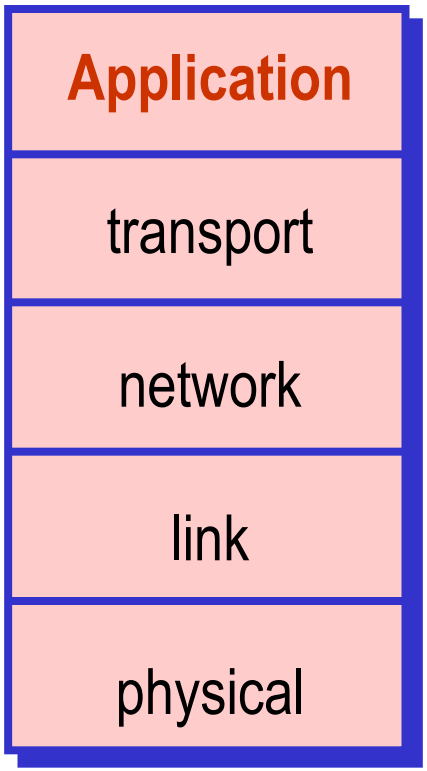
Encapsulation



Application layer

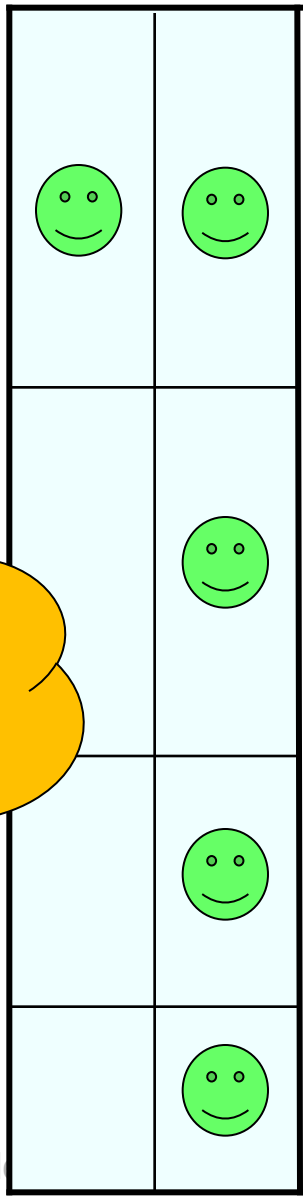


- The set of (of protocols that supports) applications that use the network
  - telnet, http, smtp, dns, ping, tracert, …
- Doesn't provide services to any other layer



Transport layer services

- Multiplexing/demultiplexing
  - adds an application-specific identifier (*port number)* so that receiving end-system can hand in incoming packet to the correct application
- Error control + *in-order* byte-stream data transfer
  - message will reach destination despite packet loss, [illegible] duplication
  - retransmit lost packets; detect, d[illegible] packets; detect and discard d[illegible]
- Flow control
  - match transmission rate to rate cu[illegible] to destination, and at the destina[illegible] itself
- Congestion control
  - senders "slow down sending rate" when network congested



What if you want only one of these services ?

Transport layer

Goal: data transfer between applications

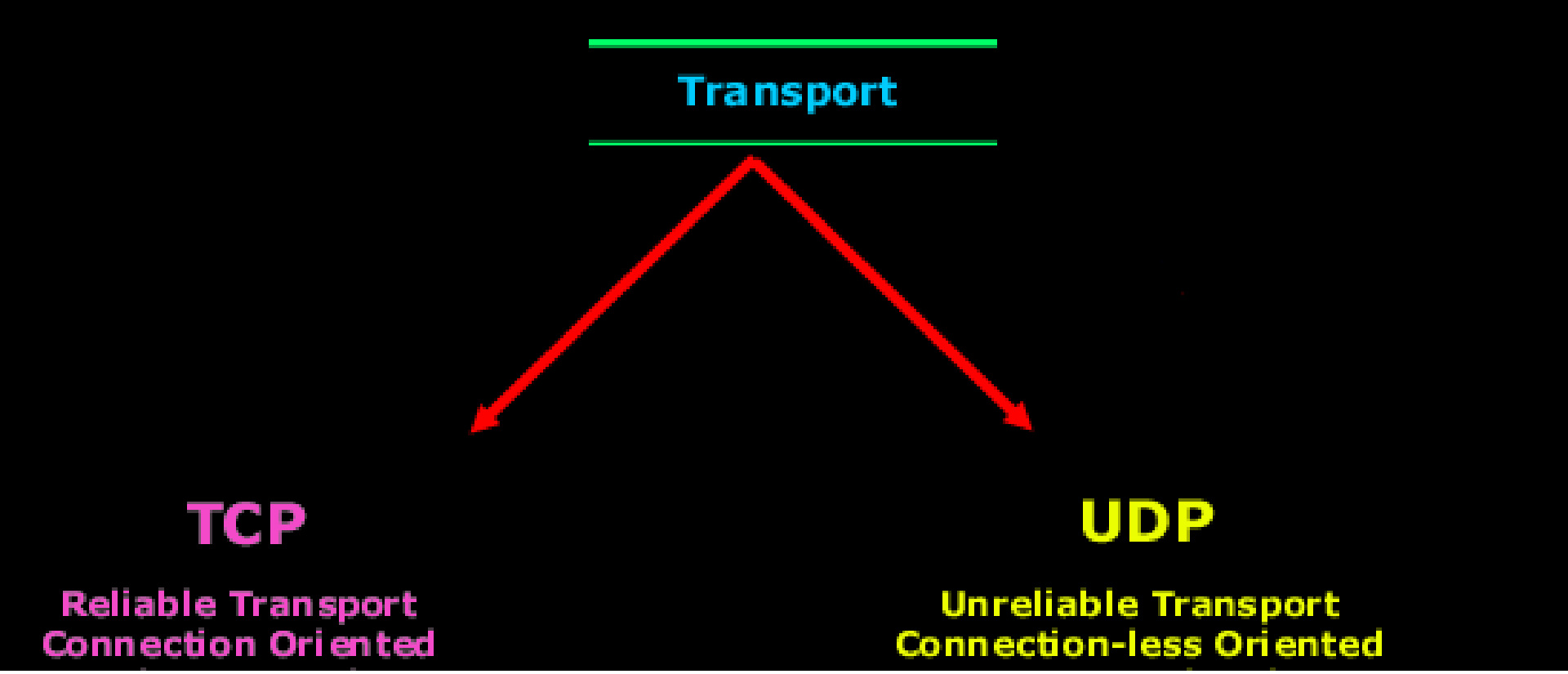
- TCP -Transmission Control RFC 793: Internet's connection-oriented service
  - *reliable, in-order* byte-stream data transfer
  - flow control
  - congestion control

- UDP - User Datagram Protocol RFC 768: Internet's connectionless service
  - unreliable data transfer
  - no flow control
  - no congestion control

RFC = Request For Comments

November 7, 2024: # of RFC's 9687

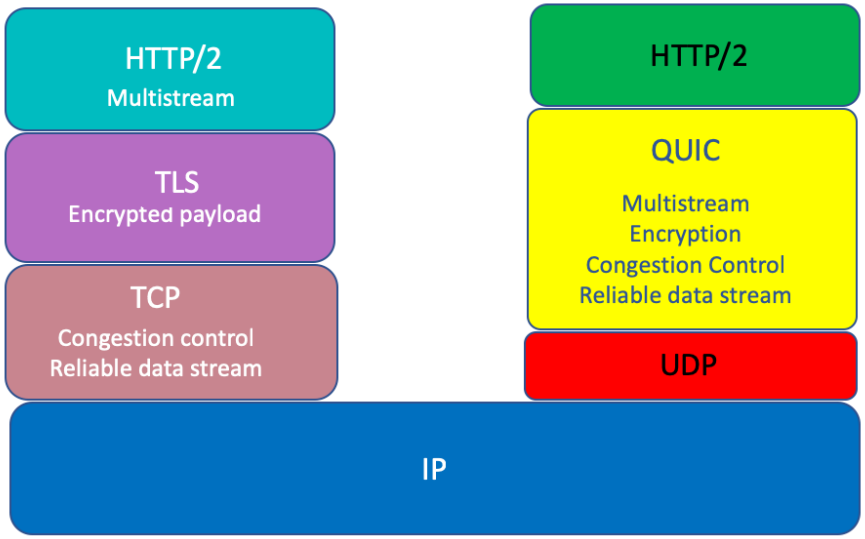
(search "RFC Index"")



HTTP (WWW)
FTP (file transfer)
Telnet (remote login)
SMTP (email)

Live streaming media (voice, video)
Internet telephony

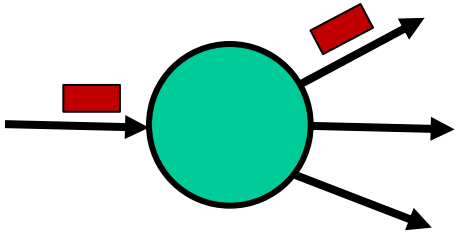
QUIC: Quick UDP Internet Connections protocol



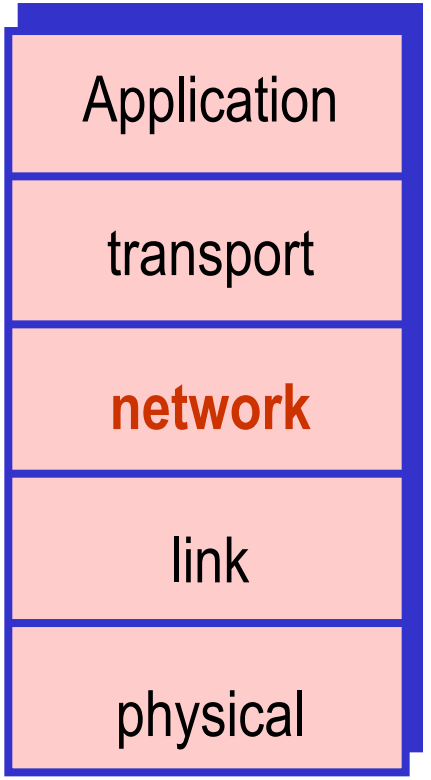
QUIC is currently developed to provide a UDP-based secure and reliable transport including loss recovery and congestion control. QUIC operates entirely in user-space.

- As of October 2024, 8.2% of all websites use QUIC.
- It's on by default in Google Chrome (so when you see YouTube …) .
- Most of Facebook's traffic now uses QUIC.
- Google applications such as YouTube utilize QUIC for Android devices since July 2016.
- RFC 9000 (May 2021)

Network layer

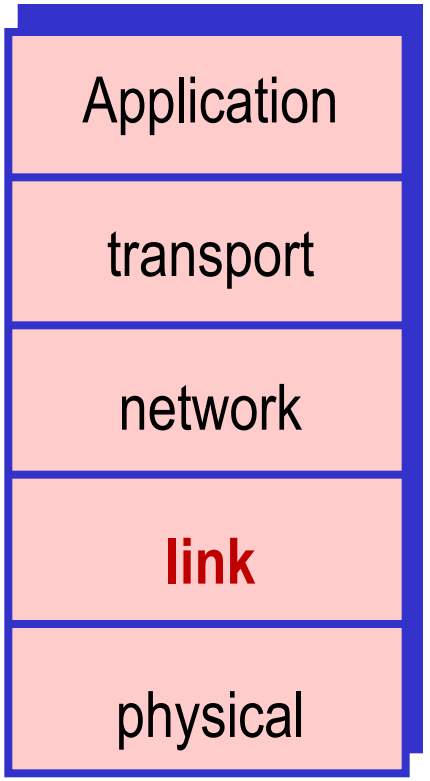


- Carries data from source to destination
- network layer is provided by Internet Protocol (IP)
- found in all end-systems and routers
- provides abstraction of end-to-end link
- segmentation and reassembly
- packet-forwarding, routing, scheduling
- unique IP addresses
- only best-effort service
- chooses which packets to drop



Link layer

- Reliable communication over a single link
- Introduces the notion of a *frame*
  - set of bits that belong together
- Internet
  - a variety protocols most common is Ethernet
  - others are WiFi, FDDI, SONET, HDLC
- Dependent on underlying physical link properties

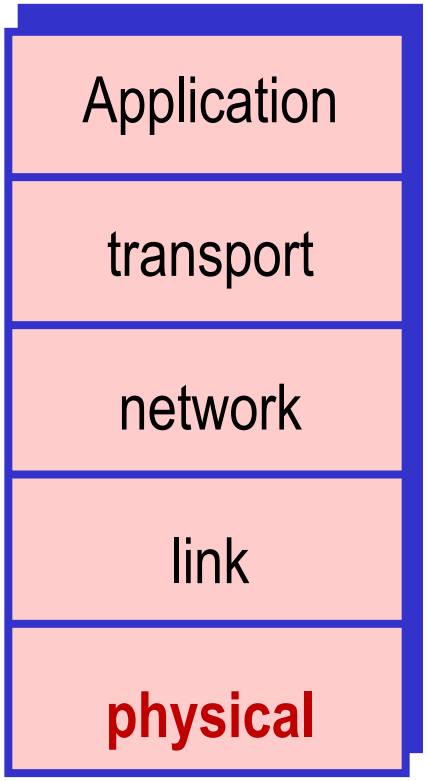




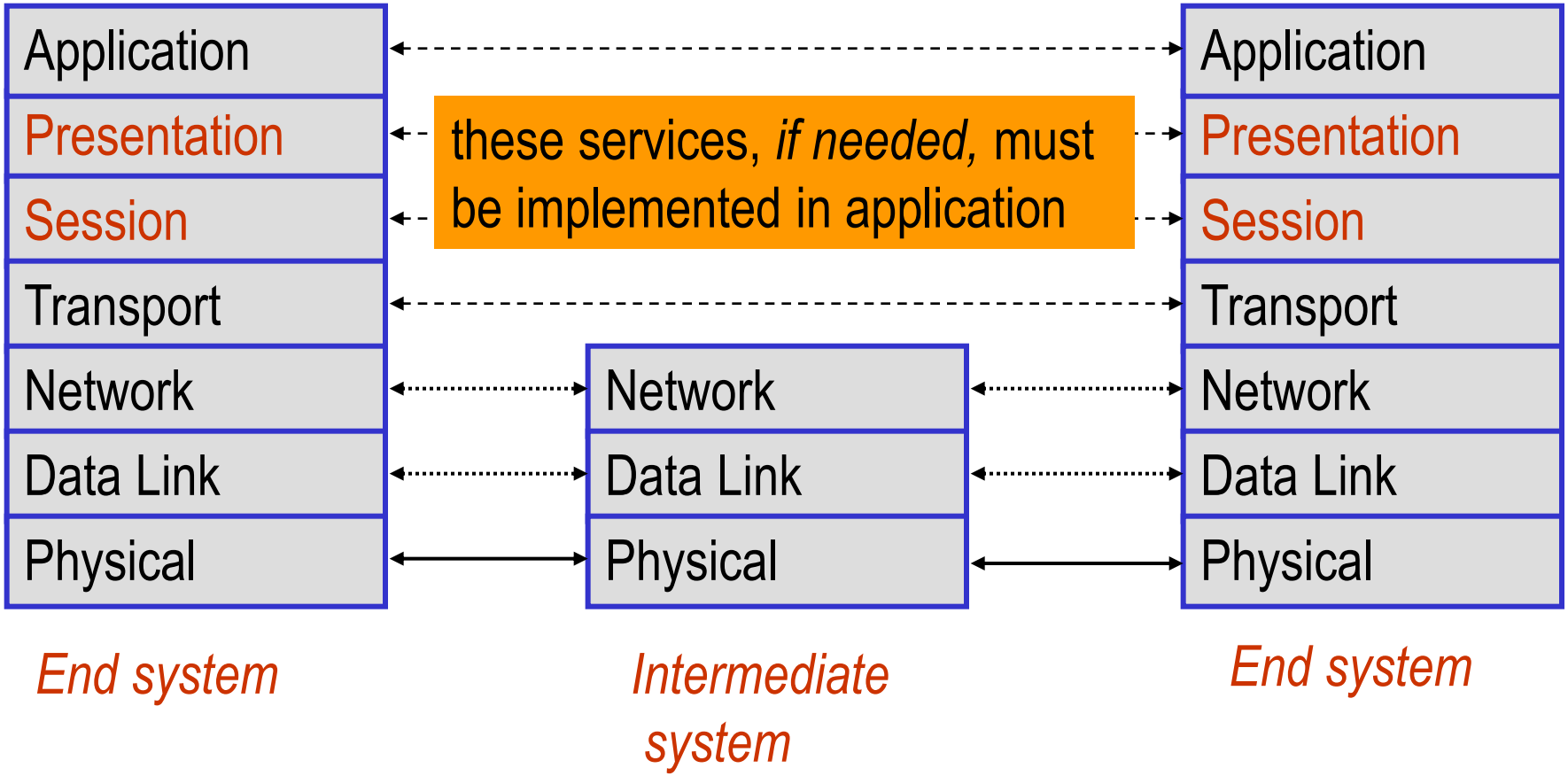
Physical layer



- Moves bits between physically connected end-systems
- Coding scheme to represent a bit
- Internet: technology to move bits on a wire, wireless link, satellite channel etc.



The OSI seven layers

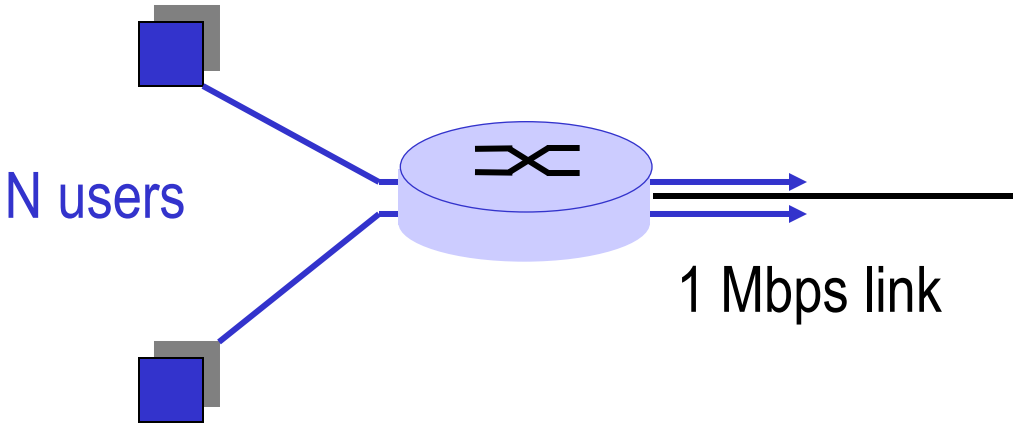


-- Open System Interconnect (OSI)

Packet switching versus circuit switching

Packet switching allows more users to use network!

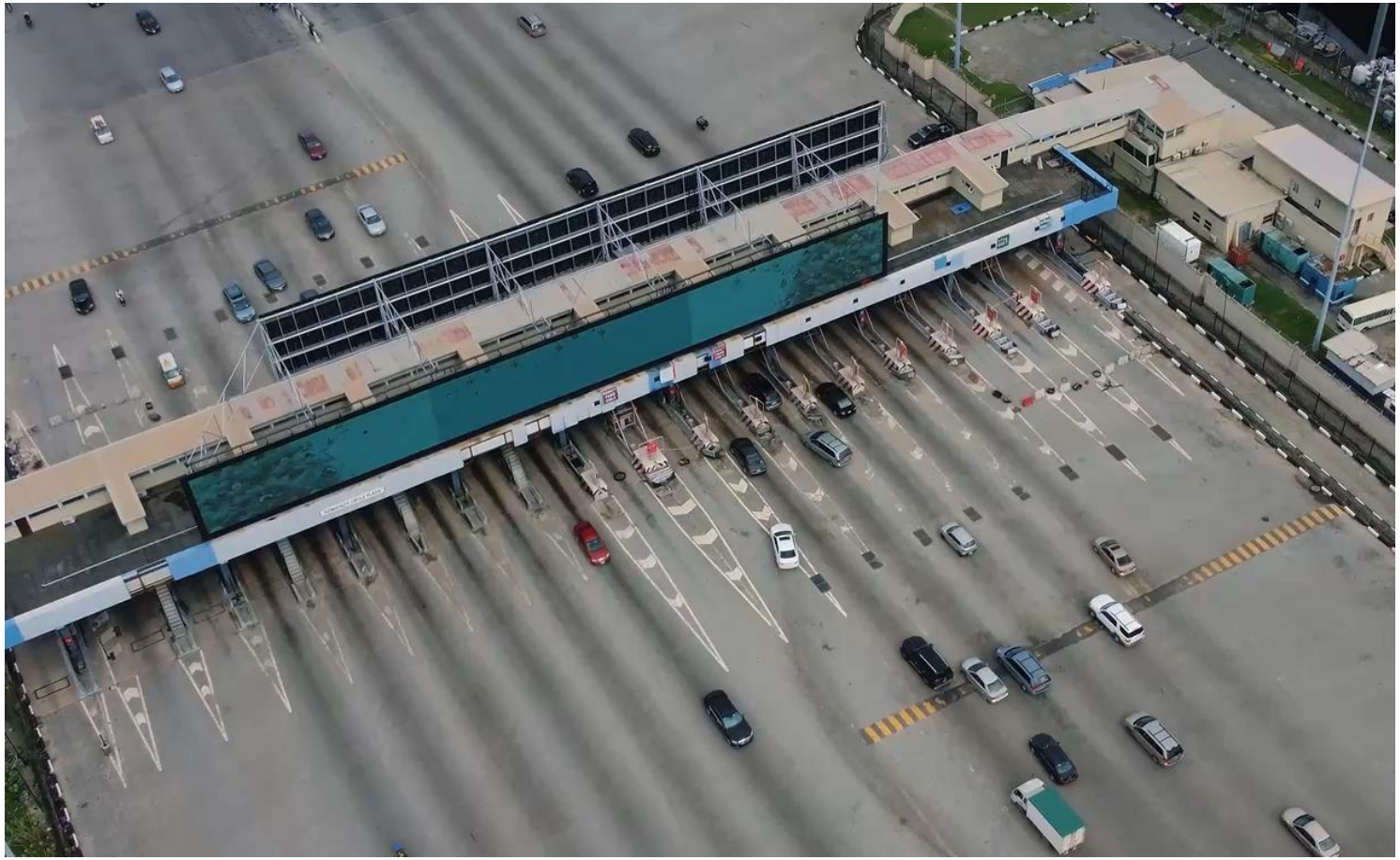
- 1 Mbit link
- each user:
  - 100 kbps when “active”
  - active 10% of time
- circuit-switching:
  - 10 users
- packet switching:
  - with 35 users, probability that there are more than 10 active users is less than .0004



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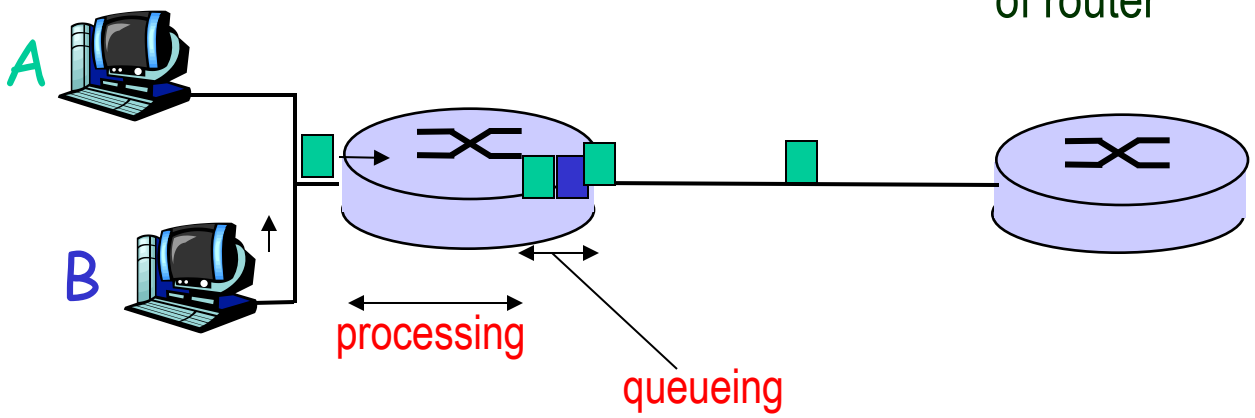
Delays in Packet Switching



Delay in packet-switched networks

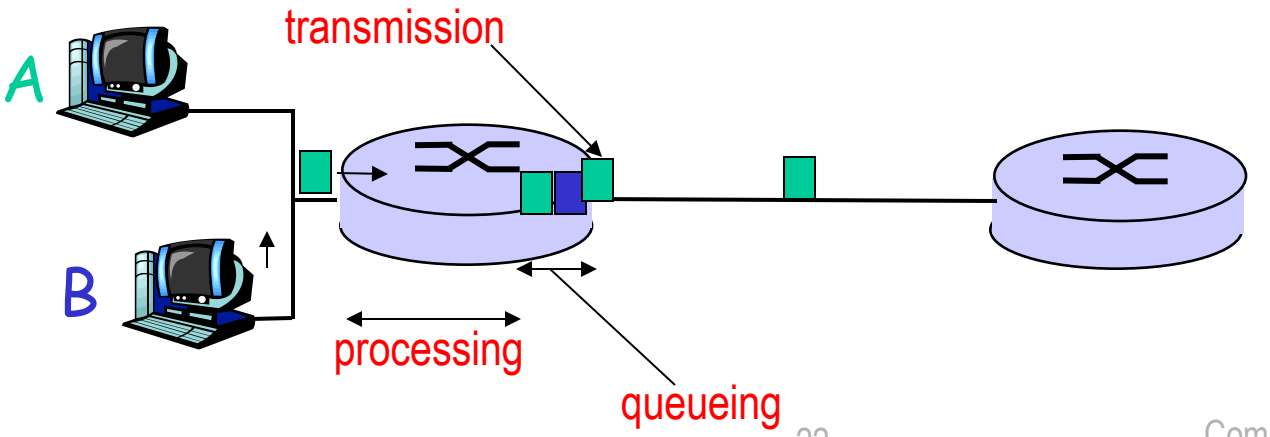
- Packets experience delay on end-to-end path
- Four sources of delay at each hop

1. Processing
   - check bit errors
   - determine output link
2. Queueing
   - time waiting at output link for transmission
   - depends on congestion level of router



Delay in packet-switched networks

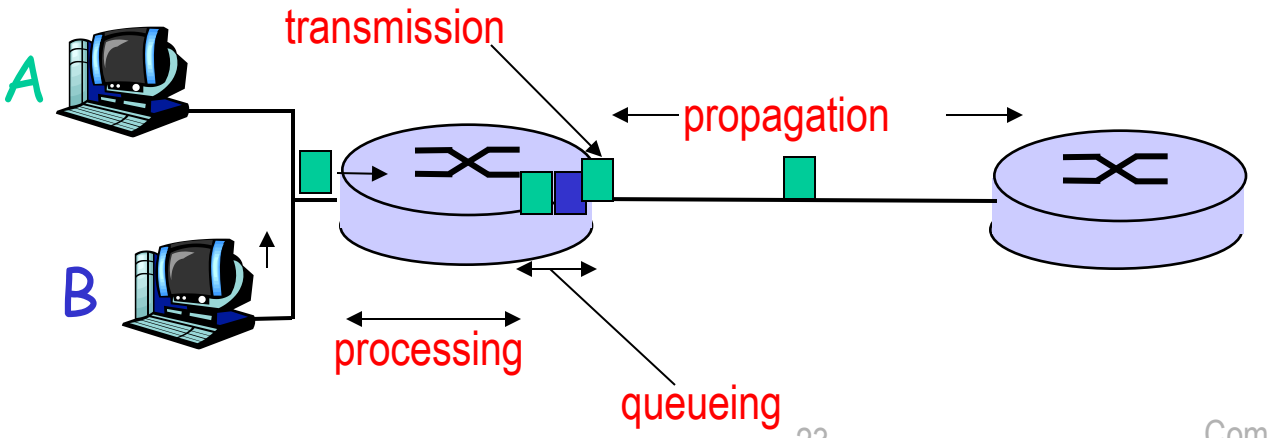
3. Transmission delay
   - R=link bandwidth (bps)
   - L=packet length (bits)
   - time to send bits into link = L/R
   - (order of microseconds)



Delay in packet-switched networks

3. Transmission delay
   - R=link bandwidth (bps)
   - L=packet length (bits)
   - time to send bits into link = L/R
   - (order of microseconds)

4. Propagation delay
   - d = length of physical link
   - s = propagation speed in medium (~$2x10^8$ m/sec)
   - propagation delay = d/s

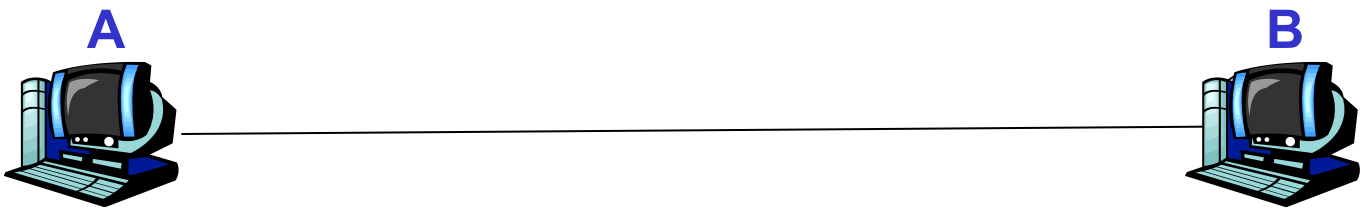


Question

-- transmission rate of 1 Mbps

-- distance is 100,000 **km** = 10^8 meter

-- propagation speed is $2.5*10^8$ meter/sec

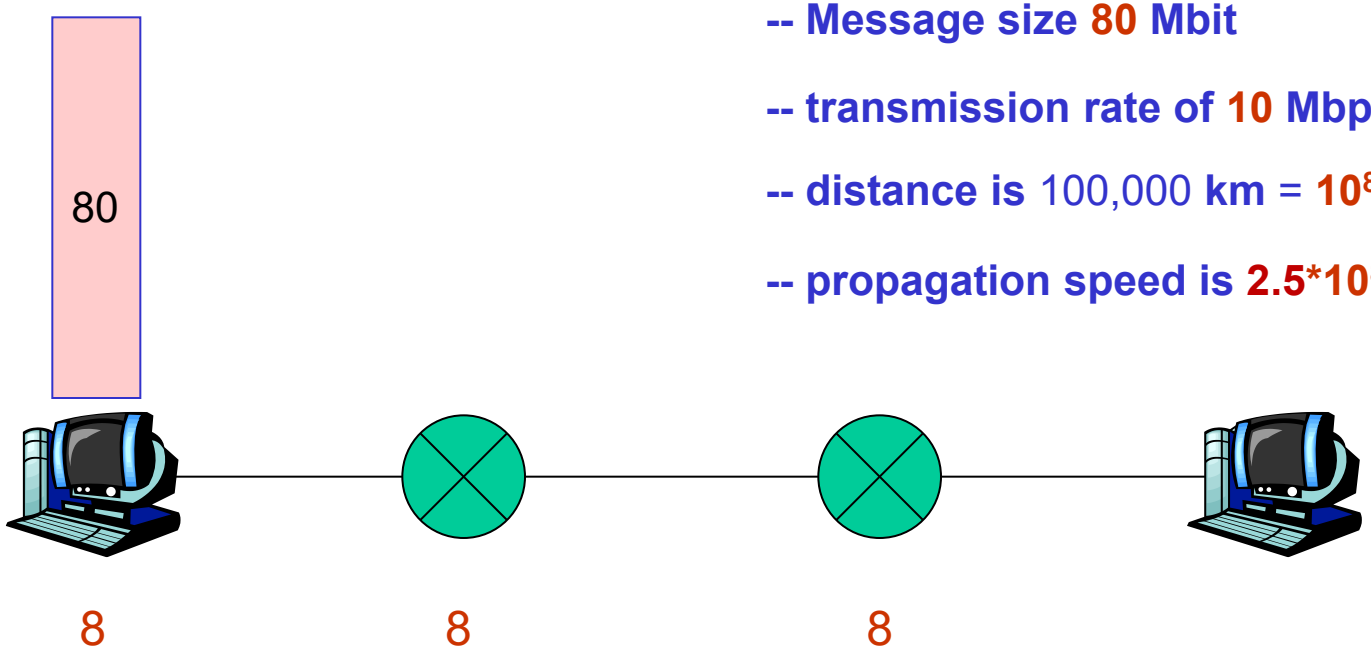


What is the propagation delay ?

Consider sending a file of 532,000 bits from host A to B. Suppose the file is sent continuously as one big message. What is the maximum number of bits that will be in the link at any given time?

What is the width (in meters) of a bit in the link ?

Question



-- Message size 80 Mbit

-- transmission rate of 10 Mbps

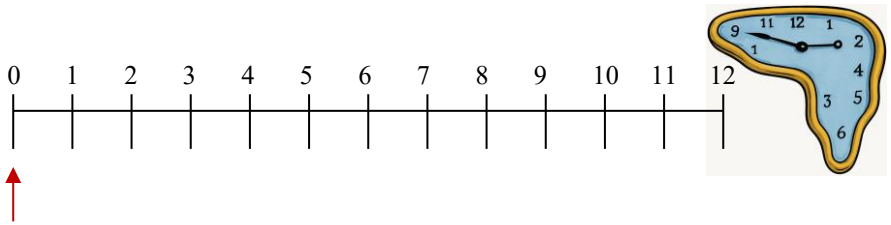
-- distance is 100,000 **km** = 10^8 meter

-- propagation speed is $2.5*10^8$ meter/sec

propagation delay = 10^8 / $2.5*10^8$ = 0.4

24.4

Question

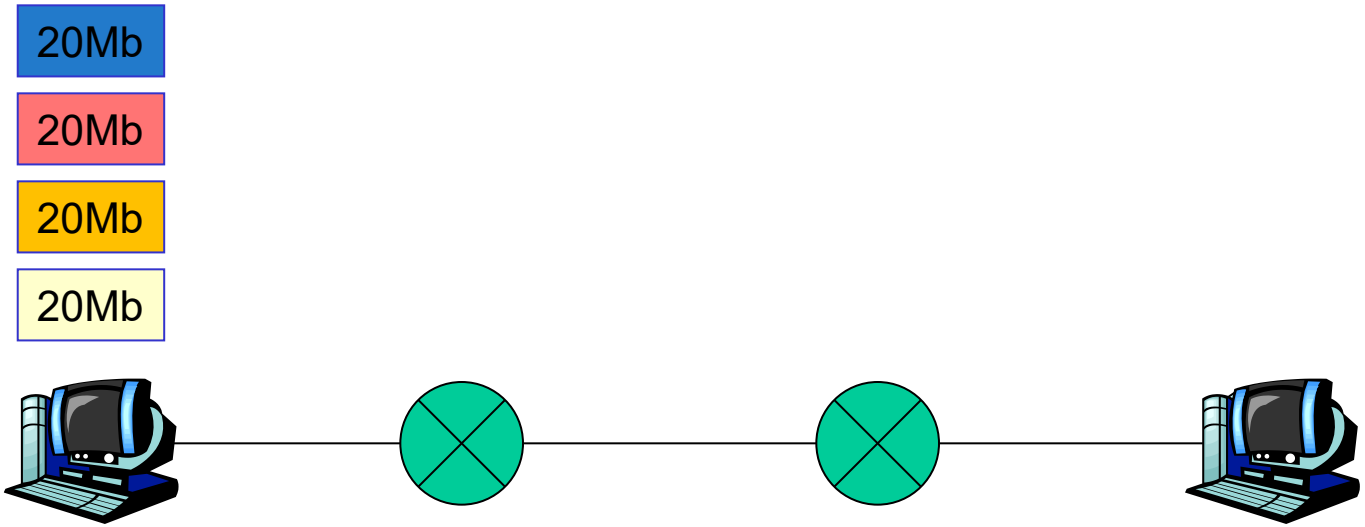


-- **Message size 80 Mb**

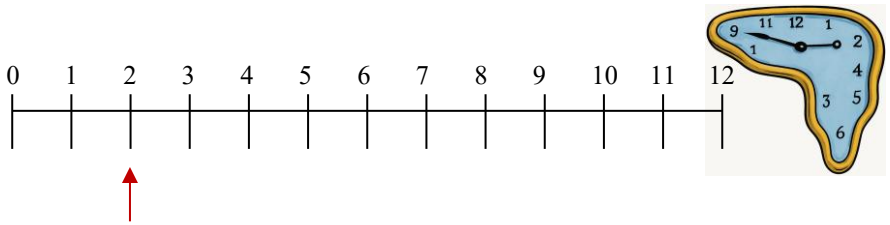
-- **transmission rate of 10 Mbps**

-- **distance is** 100,000 **km** = 10^8 meter

-- **propagation speed is** $2.5*10^8$ meter/sec



Question

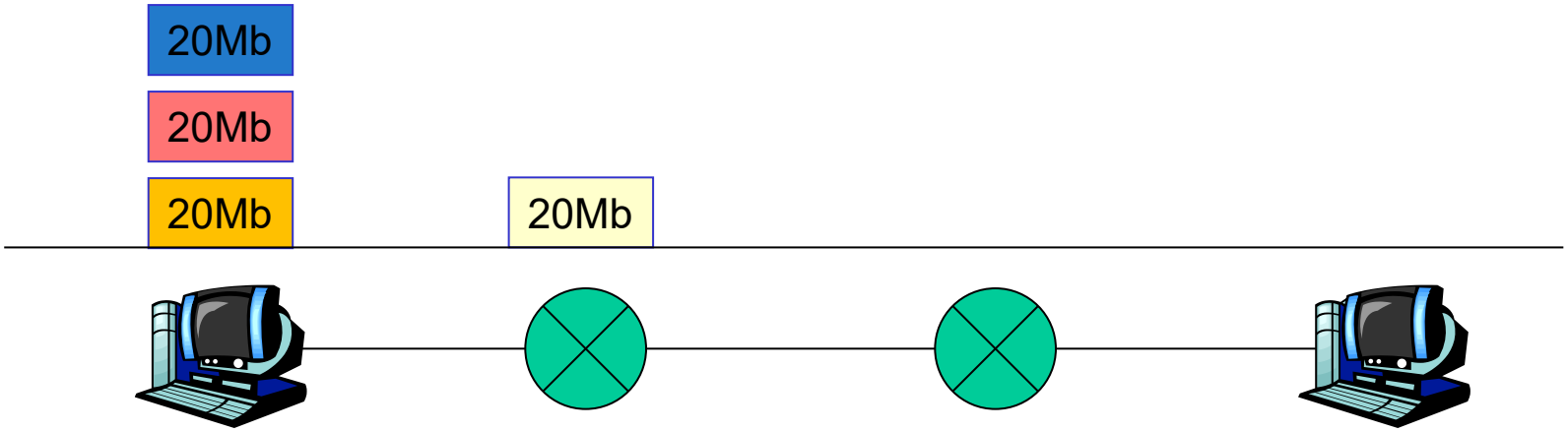


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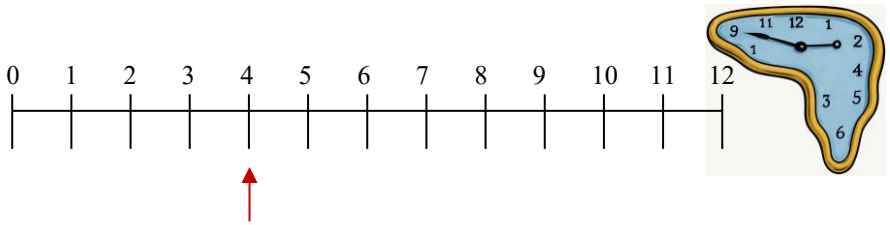
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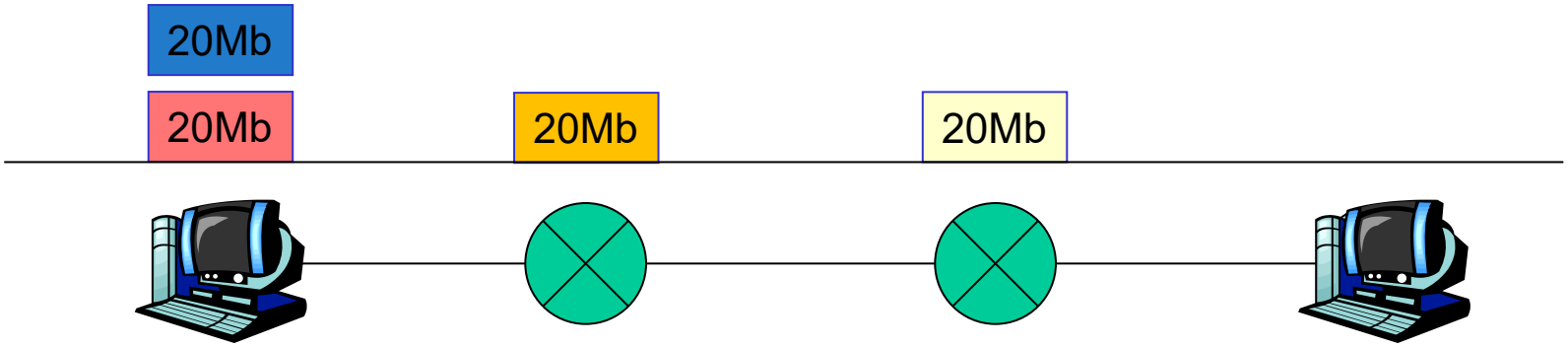


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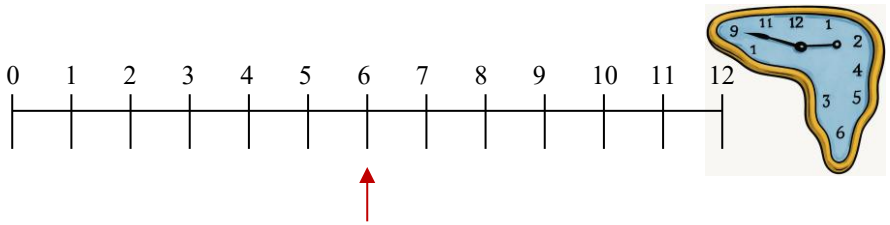
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Question



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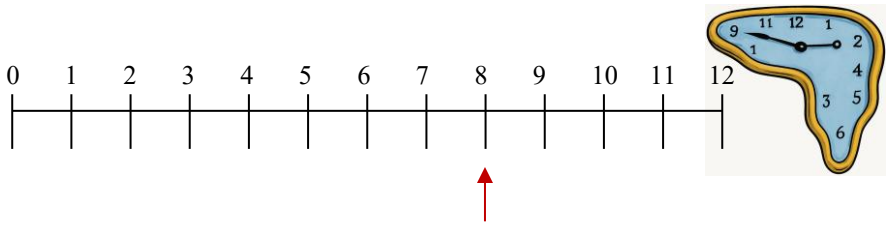
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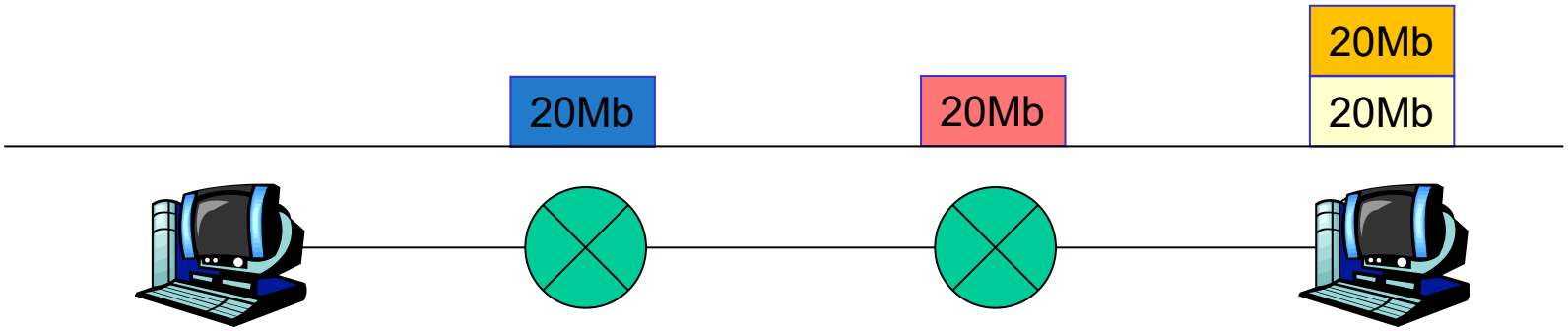


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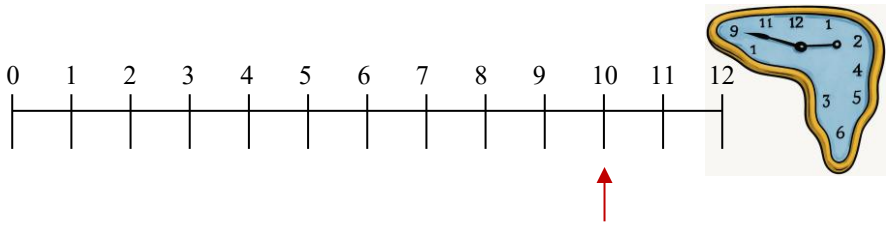
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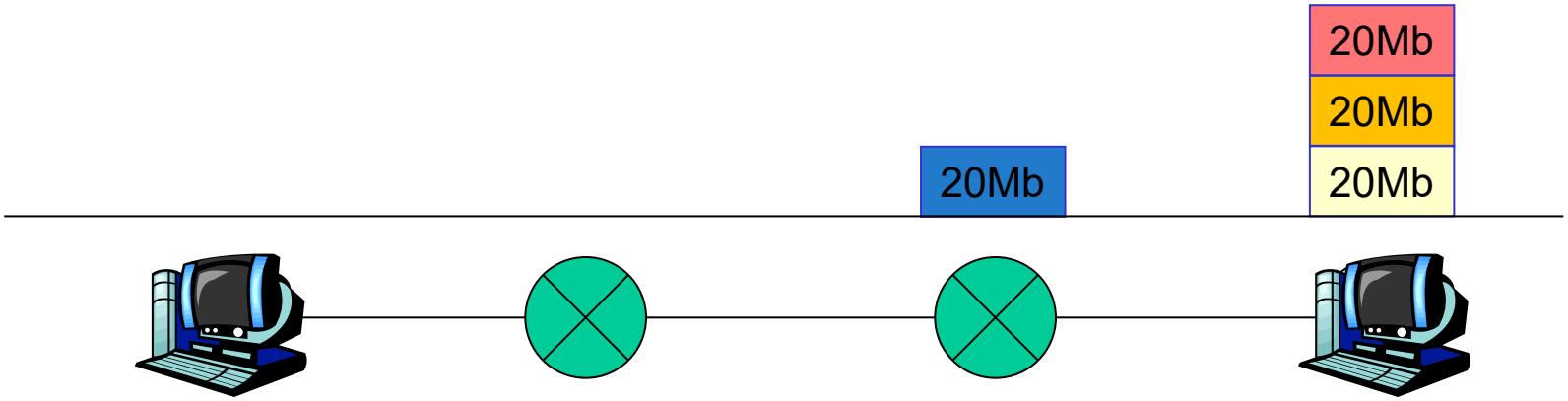


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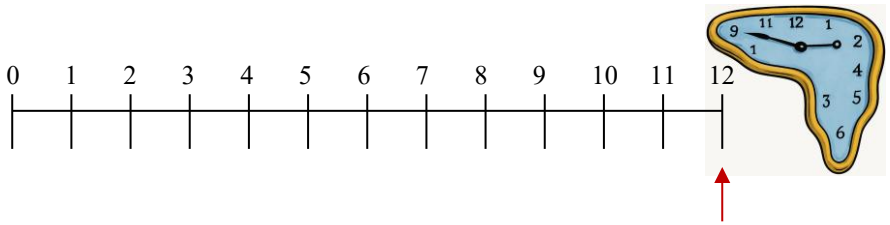
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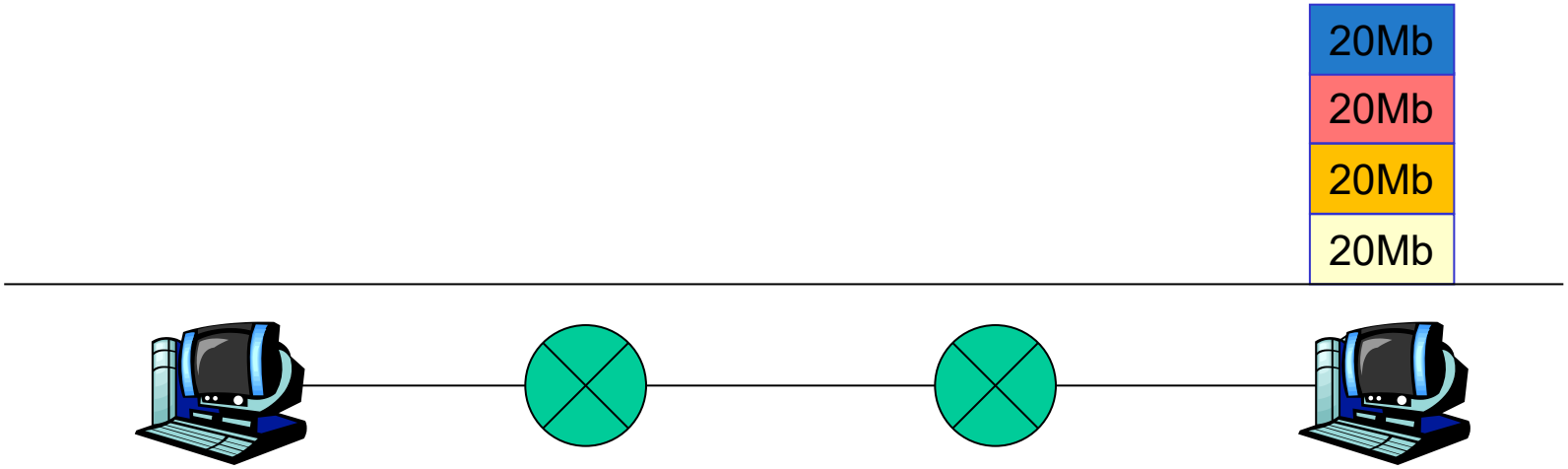


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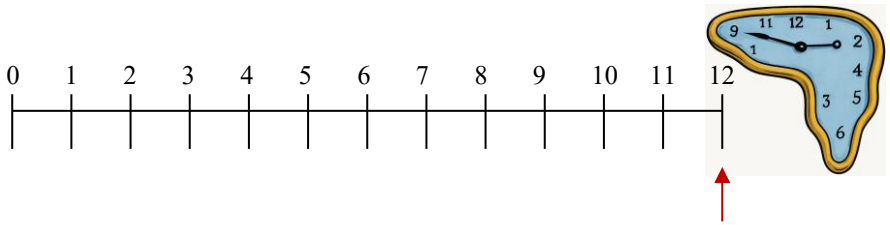
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Question



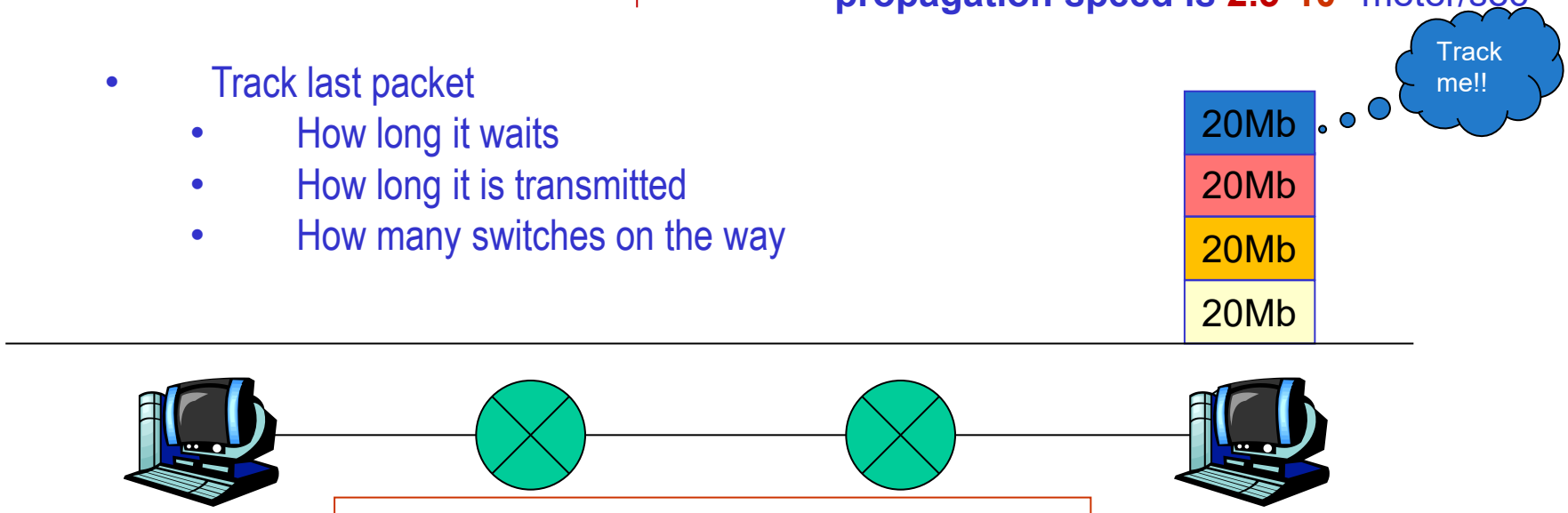
-- Message size 80 Mb

-- transmission rate of 10 Mbps

-- distance is 100,000 **km** = **10^8** meter

-- propagation speed is $2.5*10^8$ meter/sec

- Track last packet
  - How long it waits
  - How long it is transmitted
  - How many switches on the way



propagation delay = 10^8 /$2.5*10^8$ = 0.4

12.4

Total delay

$$total\ delay = d_{queue} + d_{proc} + d_{trans} + d_{prop}$$

d_{queue} = queuing delay

- depends on congestion

d_{proc} = processing delay

- typically a few microsecs or less

d_{trans} = transmission delay

- = L/R, significant for low-speed links

d_{prop} = propagation delay

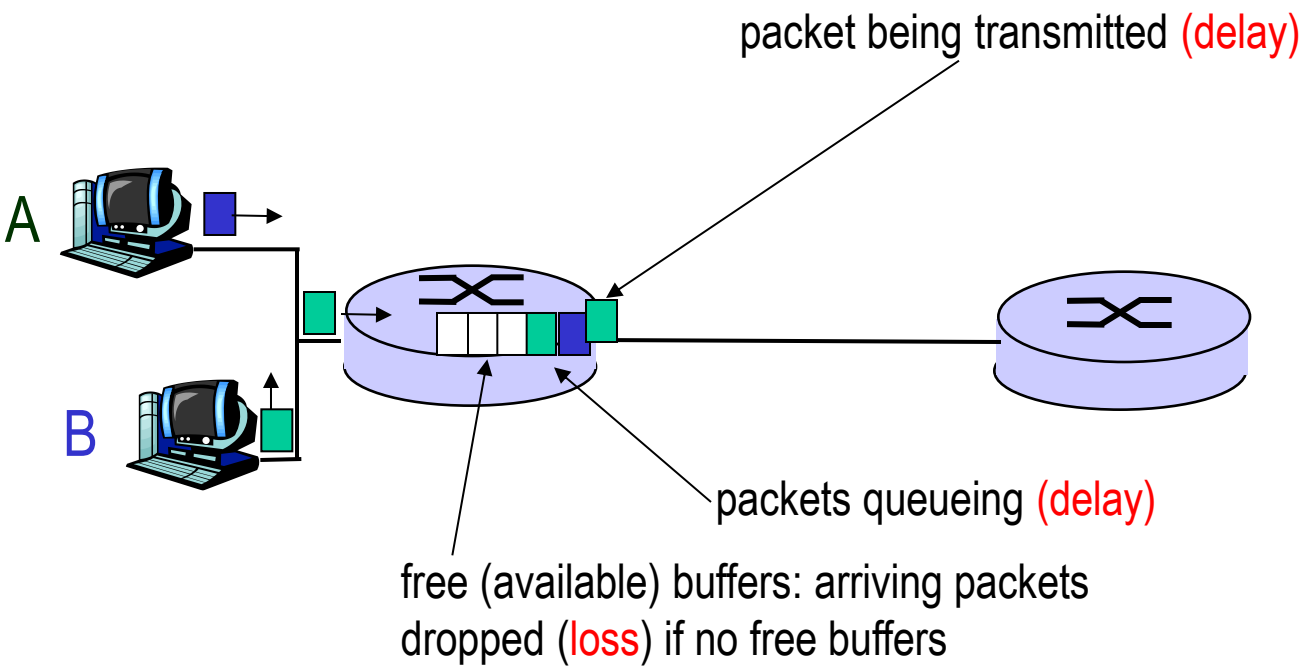
- a few microsecs

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Packet loss

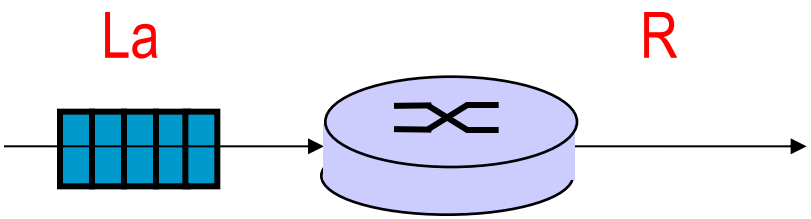
- buffer preceding link has finite capacity
- when packet arrives to full queue, packet is dropped (lost)
- lost packet may be retransmitted



Queueing delay (revisited)

- R= link bandwidth (bits/sec)
- L= packet length (bits)
- a= average packet arrival rate (packets/sec)

traffic intensity = L·a/R



- L·a/R ~ 0: average queueing delay small
- L·a/R ➔ 1: delays become large
- L·a/R > 1: more "work" arriving than can be serviced, average delay infinite!



La/R ~ 0



Today

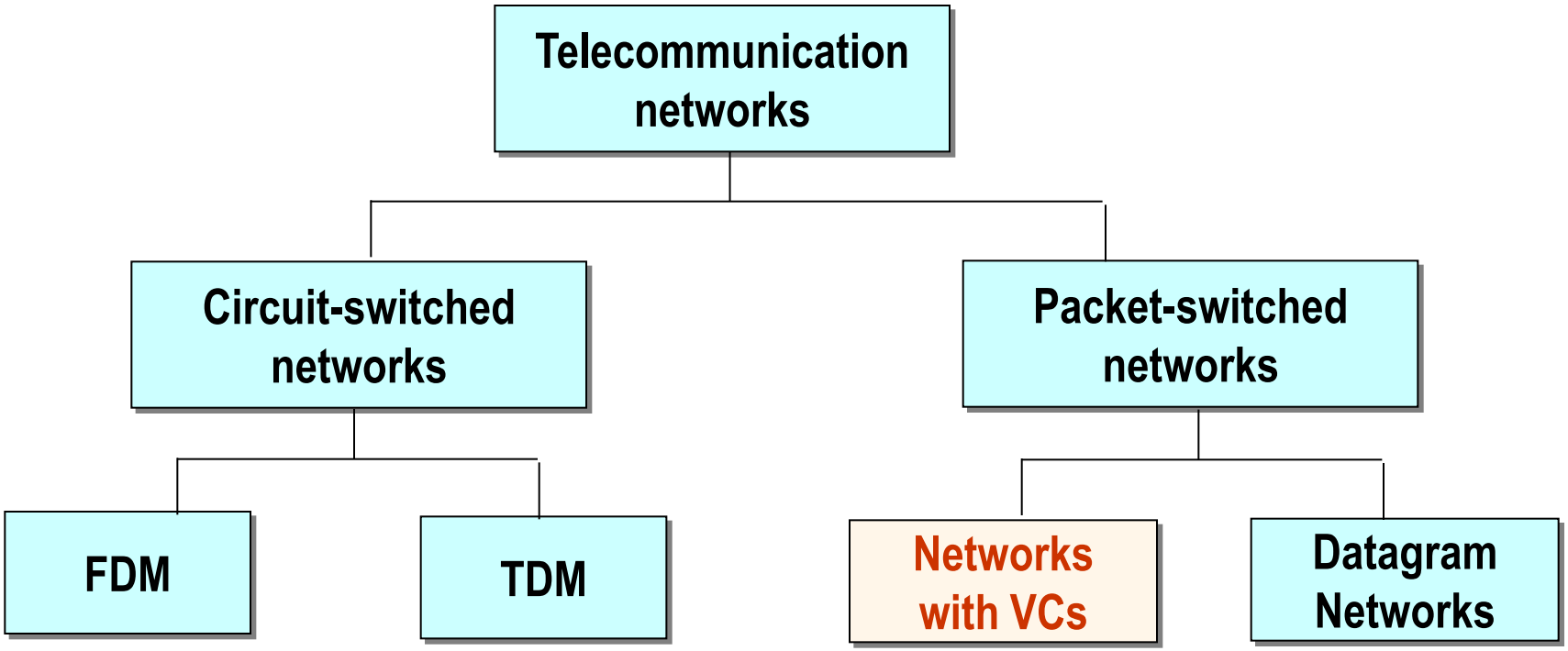
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  - Transport Layer
  - Network Layer
  - Link Layer
  - Physical Layer
- ✓ Delays in packet switching
- ✓ Packet loss
- Virtual Circuit
- Creating a network-application

Is datagram network a “winner”?

- Great for bursty data
  - resource sharing
  - no call setup
- Excessive congestion: packet delay and loss
  - protocols needed for reliable data transfer, congestion control

- Q: How to provide circuit-like behavior?
  - bandwidth guarantees needed for audio/video apps might still be a problem.

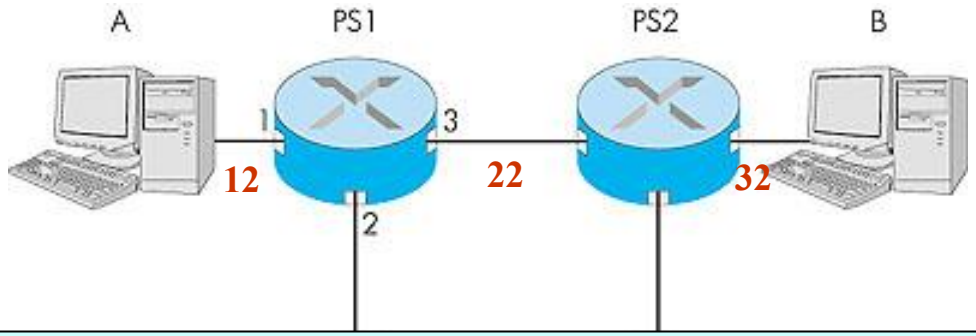
Network Taxonomy



Comparing com types

Aspect	Circuit Switching	Packet Switching	Packet Switching (Virtual Circuits)
Connection Setup	Requires dedicated path setup before transmission.	No setup required; packets sent immediately.	Setup phase required for virtual path.
Resource Usage	Resources reserved for full duration of call.	Shared dynamically among users (efficient).	Resources managed dynamically per virtual path.
Setup Delay	High – path must be established.	Low – no setup needed.	Medium – short setup phase.
Transmission Delay	Constant and predictable.	Variable – packets may take different routes.	Fairly constant – all packets use same virtual path.
Reliability & Order	Guaranteed in-order delivery.	Possible loss or reordering of packets.	Order preserved like in circuit switching.
Bandwidth Utilization	Inefficient – bandwidth held even when idle.	Very efficient – used only when data sent.	Moderately efficient – mix of stability and sharing.
Congestion Handling	Difficult to reroute during congestion.	Highly flexible – packets rerouted easily.	Moderate – rerouting possible but complex.
Real-Time Suitability	Very suitable (e.g., traditional telephony).	Less suitable – variable delays.	Fairly suitable – more consistent delays.

Virtual circuit networks



Why not using one CV# (tag) for the whole route?

PS3 PS4

Incoming interface	Incoming VC #	Outgoing interface	Outgoing VC #
1	12	3	22
2	63	1	18
3	97	2	17
1	97	3	87
…	…	…	…

Packet-switched networks: routing

Goal: move packets among routers from source to destination

Virtual Circuit network

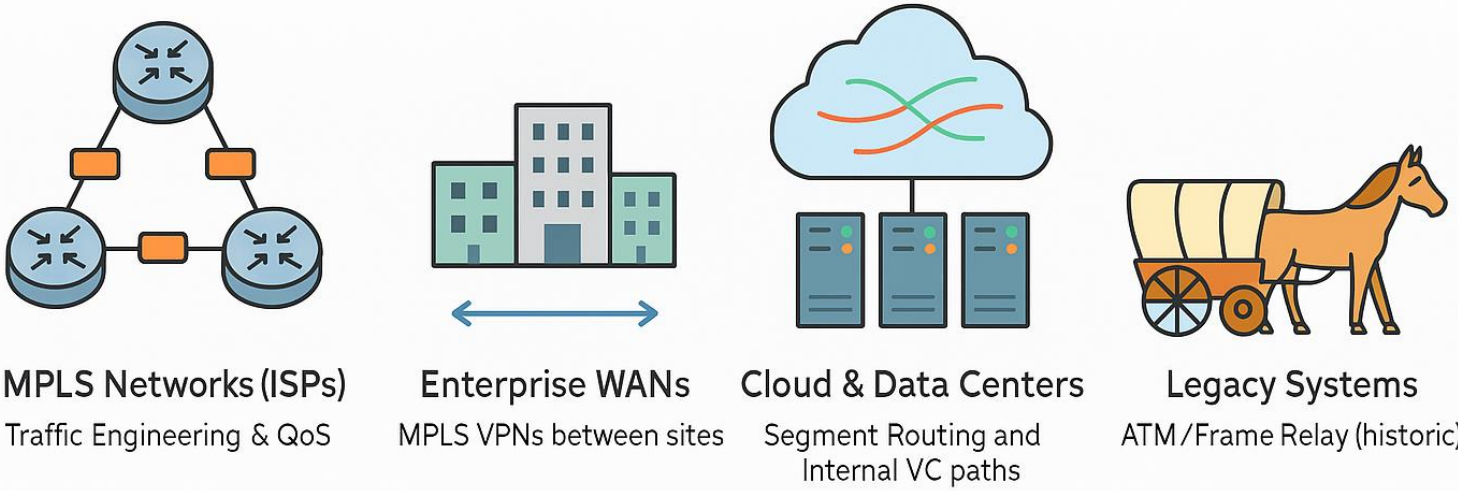
- each packet carries tag (virtual circuit ID), tag determines next hop
- fixed path determined at *call setup time*, remains fixed thru call
- routers maintain per-call state. This is expensive and complex.

Datagram network

- *destination address* determines next hop
- routes may change during session
- analogy: driving, asking directions

*An example of packet-switching technology that uses virtual circuit is **ATM** (**A**synchronous **T**ransfer **M**ode)*

Where will we find VC-sw



MPLS Networks (ISPs)
Traffic Engineering & QoS

Enterprise WANs
MPLS VPNs between sites

Cloud & Data Centers
Segment Routing and Internal VC paths

Legacy Systems
ATM/Frame Relay (historic)

Today

- ✓ Last week Review
- ✓ Introducing the 5 Layers of the internet protocol – cont.
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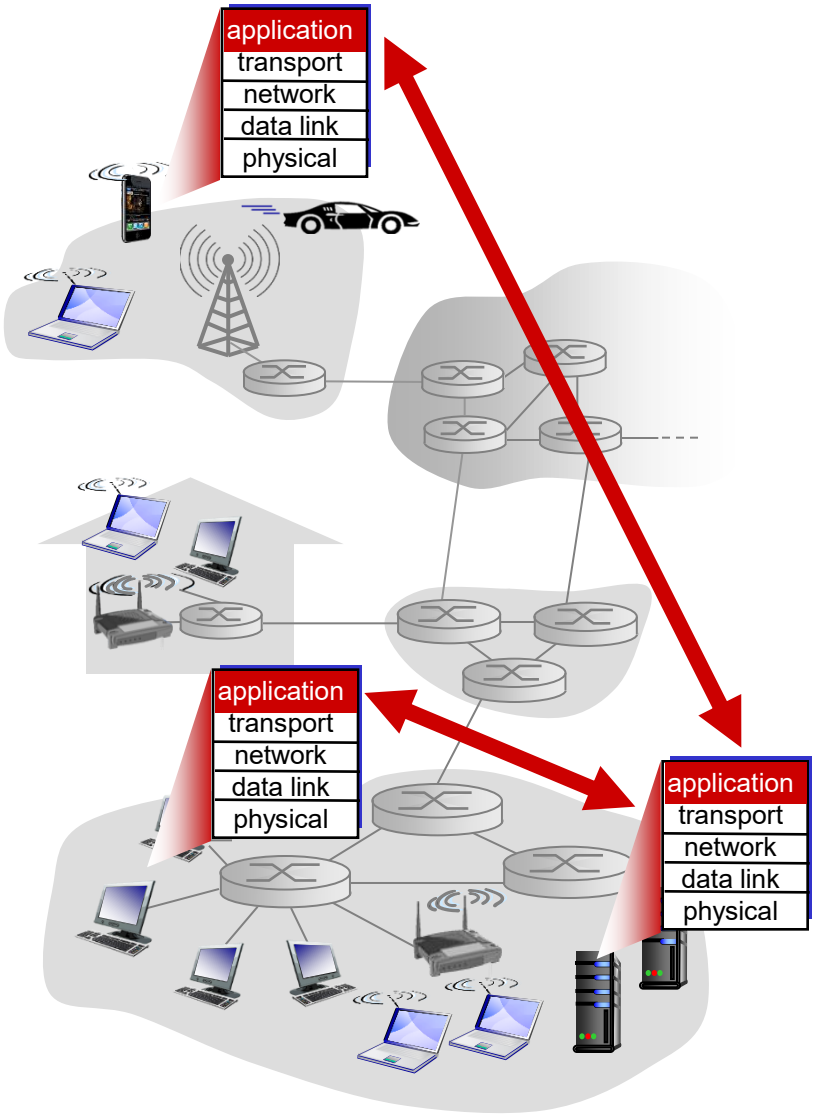
Creating a network app

Write programs that:

- run on (different) *end systems*
- communicate over network
- e.g., web server software communicates with browser software

No need to write software for network-core devices

- network-core devices do not run user applications
- applications on end systems allows for rapid app development, propagation



Network applications: some jargon

- A process: a program that is running within a host.
- Interprocess communication: communication between processes running on the same host.
- Application-layer protocol: communication between processes running in different hosts.



Addressing Processes

- Every host has a unique 32-bit IP address
- Q: Does the IP address of the host on which the process runs suffice for identifying the process?



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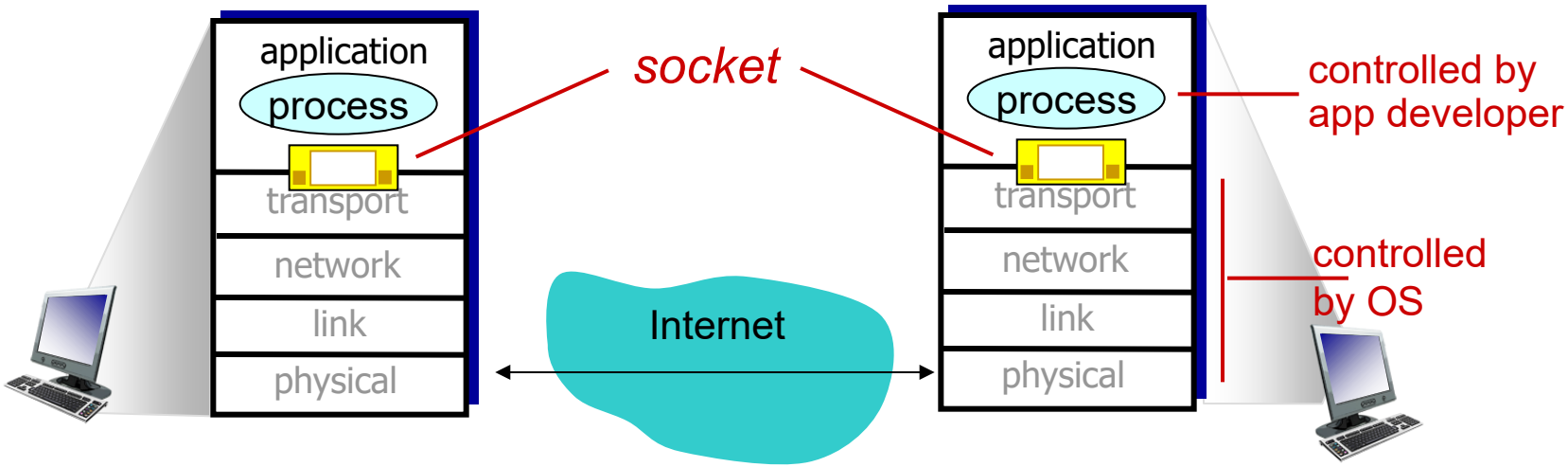
- Answer: No, many processes can be running on same host.

Socket-programming

Socket: a door between application process and end-end-transport protocol (UDP or TCP)

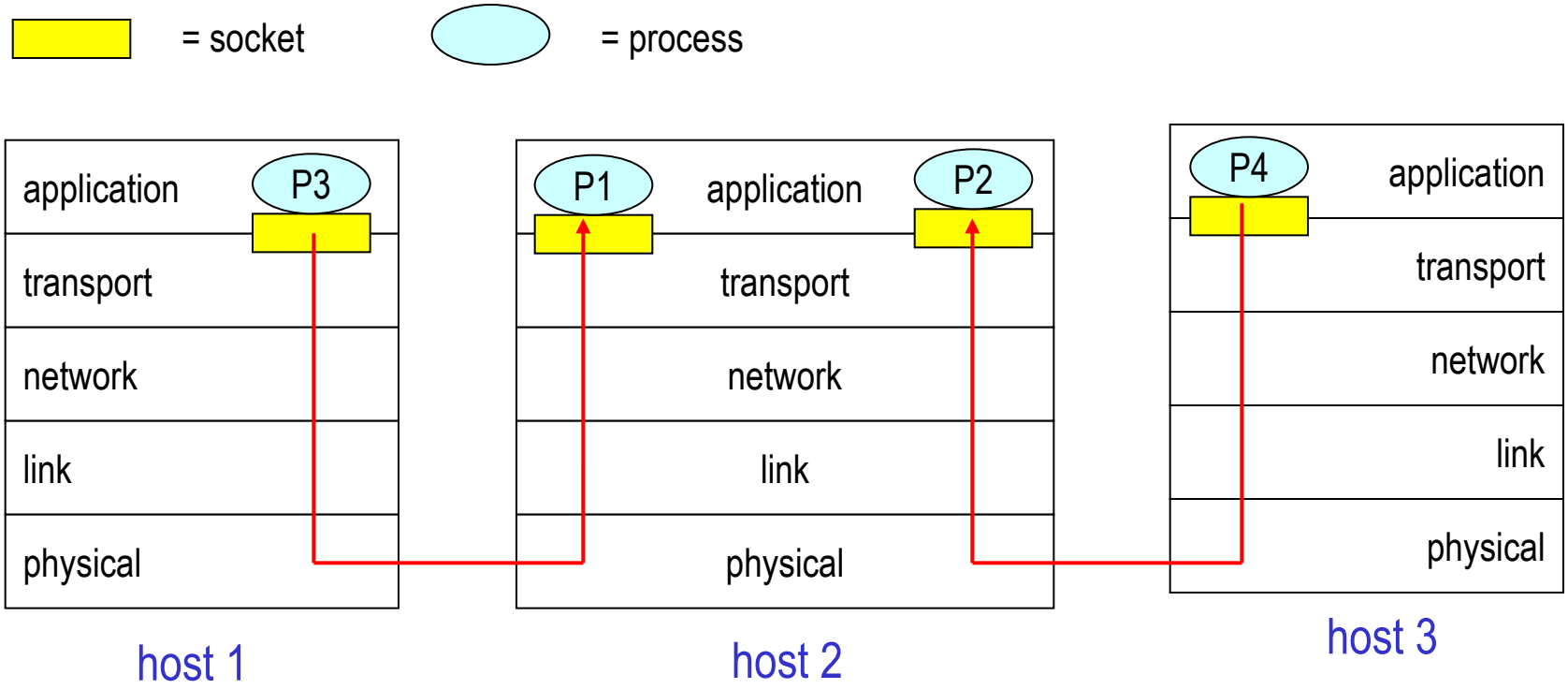
Port number: allows receiving host to determine to which local process the message should be delivered

Socket = IP address + port number



… lots more on this later

Socket-programming



Addressing Processes

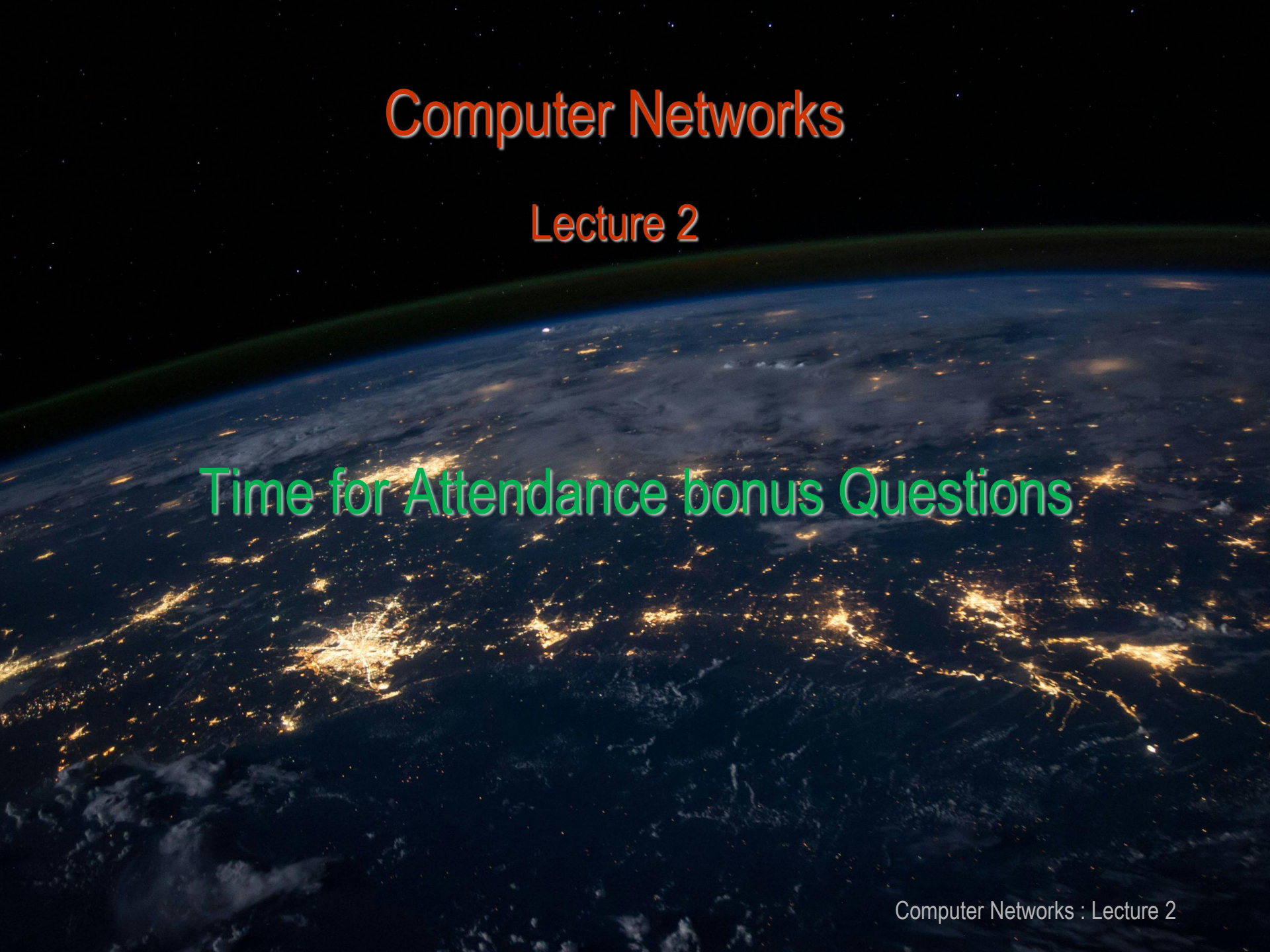
- Identifier includes both the IP address and port numbers associated with the process on the host.
- We write: www.google.com
  - Have to be translated to IP
  - Have to find a port
- IP (numbers = DNS)
- Port = Standard (0-1024)
- Actually: http://www.google.com
- Example port numbers:
  - HTTP server: 80
  - Mail server: 25



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Computer Networks

Lecture 2

Time for Attendance bonus Questions

Warriors of the Net

http://www.warriorsofthe.net/movie.html

It is recommended to see the following movie in Hebrew or English (13 min) !



Summary

Material covered:

- Internet overview
- Circuit vs. packet switching
- What's a protocol?
- network edge, core
- Backbones, NAPs, ISPs
- Performance: loss, delay
- Layering and service models

You now hopefully have:

- Context, overview, "feel" of networking
- More depth, detail *later* in course